

12 “CHANCE” FORGETTING EVENTS ON *CIFAR-10*

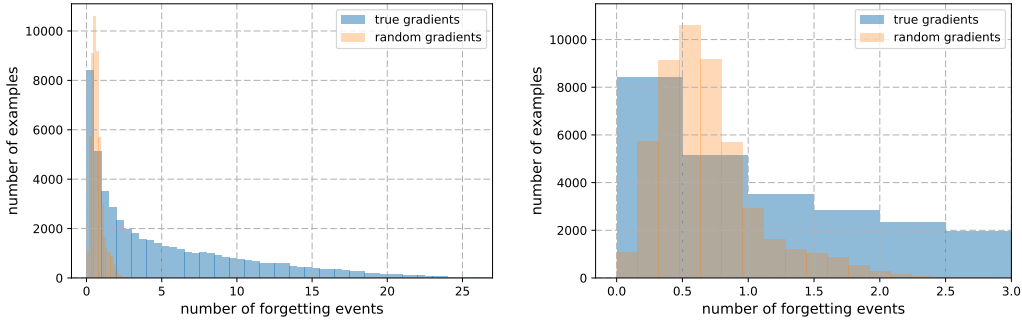


Figure 13: Histogram of forgetting events under true and random gradient steps. (Right) Zoomed-in version where the number of forgetting events is capped at 3 for visualization.

Forgetting events may happen by “chance”, i.e. some learning/forgetting events may occur even with random gradients. In order to estimate how large the effect of “chance” is, we compute the forgetting events of a classifier obtained by randomizing the update steps. To keep the statistics of the gradients similar to those encountered during SGD, we proceed as follows:

1. Before the beginning of training, clone the “base” classifier into a new “clone” classifier with the same random weights.
2. At each training step, shuffle the gradients computed on the base classifier and apply those to the clone (the base classifier is still optimized the same way): this ensures that the statistics of the random updates match the statistics of the true gradients during learning.
3. Compute the forgetting events of the clone classifier on the training set exactly as is done with the base classifier.

The results can be found in Fig 13, showing the histogram of forgetting events produced by the clone network, averaged over 5 seeds. This gives an idea of the chance forgetting rate across examples. In this setting, examples are being forgotten by chance at most twice.

13 CONFIDENCE ON FORGETTING EVENTS FOR *CIFAR-10*

In order to establish confidence intervals on the number of forgetting events, we computed them on 100 seeds and formed 20 averages over 5 seeds. In Fig 14, we show the average (in green), the bottom 2.5 percentile (in blue) and top 2.5 percentile (in orange) of those 20 curves.

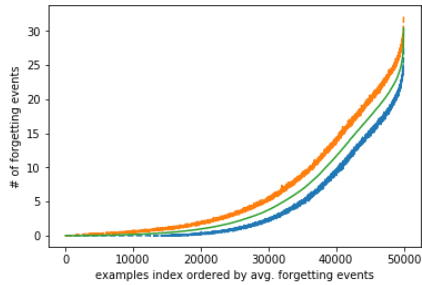


Figure 14: 95% confidence interval on forgetting events averaged over 5 seeds.

14 VISUALIZATION OF FORGETTABLE AND UNFORGETTABLE IMAGES

See Fig 15 for additional pictures of the most unforgettable and forgettable examples of every *CIFAR-10* class, when examples are sorted by number of forgetting events (ties are broken randomly).

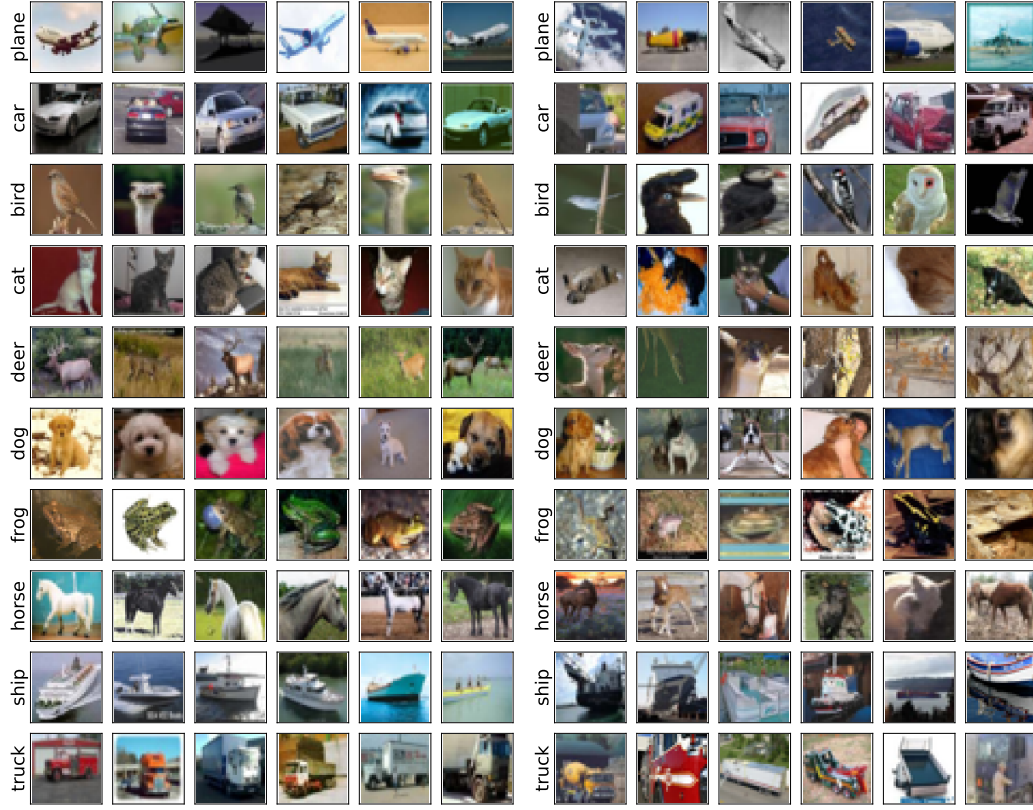


Figure 15: Additional pictures of the most unforgettable (*Left*) and forgettable examples (*Right*) of every *CIFAR-10* class, when examples are sorted by number of forgetting events (ties are broken randomly). Forgettable examples seem to exhibit peculiar or uncommon features.

15 FORGETTING IN *CIFAR-100*

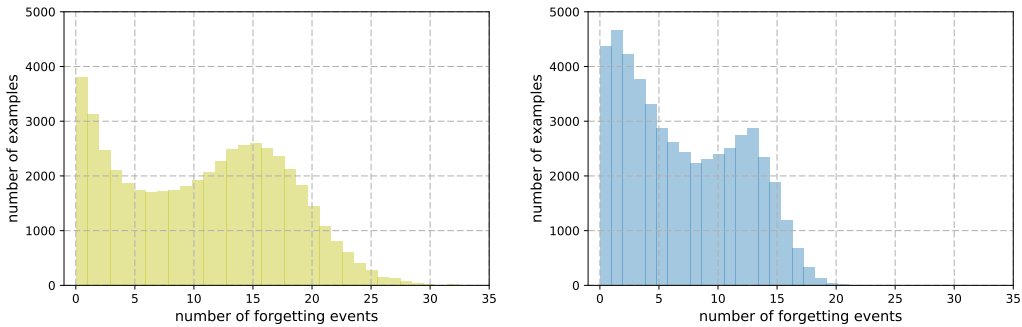


Figure 16: Left: distribution of forgetting events in *CIFAR-100*. Right: distribution of forgetting events in *CIFAR-10* when 20% of the labels are changed at random. The distribution of forgetting in *CIFAR-100* is much closer to that of forgetting in the noisy *CIFAR-10* than it is to forgetting in the original datasets presented in Figure 1.

The distribution of forgetting events in *CIFAR-100* is shown in Figure 16. There are 3809 unforgettable examples (7.62% of the training set). *CIFAR-100* is the hardest to classify out all of the presented datasets and exhibits the highest percentage of forgetting events. This finding further supports the idea that there may be a correlation between the forgetting statistics and the intrinsic

dimension of the learning problem. Additionally, each *CIFAR-100* class contains 10 times fewer examples than in *CIFAR-10* or the *MNIST* datasets, making each image all the more useful for the learning problem.

We also observe that the distribution of forgetting in *CIFAR-100* is much closer to that of forgetting in the noisy *CIFAR-10* than it is to forgetting in the original datasets presented in Figure 1. Visualizing the most forgotten examples in *CIFAR-100* revealed that *CIFAR-100* contains several images that appear multiple times in the training set under different labels. In Figure 17, we present the 36 most forgotten examples in *CIFAR-100*. Note that they are all images that appear under multiple labels (not shown: the "girl" image also appears under the label "baby", the "mouse" image also appears under "shrew", one of the 2 images of 'oak_tree' appears under 'willow_tree' and the other under 'maple_tree').

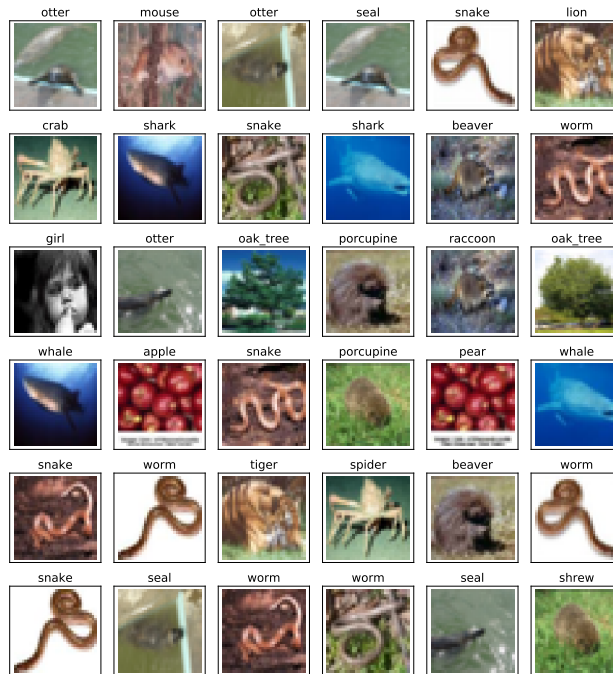


Figure 17: The 36 most forgotten examples in *CIFAR-100*. Note that they are all images that appear under multiple labels (not pictured: the "girl" image also appears under the label "baby", the "mouse" image also appears under "shrew", one of the 2 images of 'oak_tree' appears under 'willow_tree' and the other under 'maple_tree').

We perform the same removal experiments we presented in Figure 5 for *CIFAR-100*. The results are shown in Figure 18. Just like with *CIFAR-10*, we are able to remove all unforgettable examples (8% of the training set) while maintaining test performance.